

Unit 11: Transport layer - congestion control and QoS

CONTENTS

Objectives

Introduction

11.1 Data Traffic

11.2 Congestion Control

11.3 Quality of Service

11.4 Techniques to improve quality of service

11.5 Traffic Shaping

Summary

Keywords

Review Questions

Self-Assessment

Self-Assessment Answer

Further Readings

Objectives

- understand traffic descriptor and traffic profiles.
- understanding congestion control categories.
- learning flow characteristics.
- techniques to improve quality of service.
- learning traffic shaping techniques.

Introduction

So, first and foremost, congestion management and quality of service are two problems that are so closely linked that improving one automatically improves the other. You're missing one strategy if you're ignoring the other as well. As a result, most strategies for preventing or eliminating congestion in a network often increase network quality of operation. A major target of congestion management and quality of service is data traffic. In congestion control, we aim to reduce traffic congestion, and in quality of service, we strive to build an optimal atmosphere for traffic.

11.1 Data Traffic

But, before we get into congestion management and service efficiency, let's take a look at data flow. So, under the heading of data flow, we'll talk about two terms:

- traffic descriptors and
- traffic profiles.

Traffic Descriptor

You can see the data rate, absolute burst size peak rate, and average data rate in the Figure 1 below. What does the information imply? The average data rate is the number of bits transmitted over time divided by the number of seconds in that time, indicating how much data you will transmit in a given second. The average data rate is a very helpful traffic characteristic because it shows how much bandwidth the traffic requires on average.

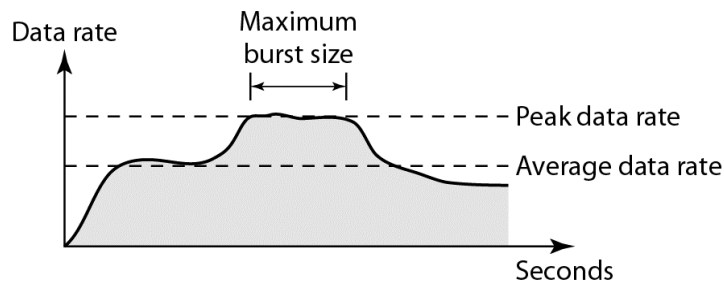


Figure 1 Traffic Descriptors

The maximum y axis value, the peak data rate, is a very significant metric, as you can see in the graph. Peak data rate indicates that there is a sudden increase in traffic, which is why the bar is rising. Peak data rate indicates that there was a lot of traffic at the time, and burst duration indicates how long the traffic was at its peak. As a result, peak data rate is a critical metric since it shows the network's maximum bandwidth, which is needed for traffic to flow without interruption. So, although the peak data rate is certainly a key advantage for the network, the actual burst size should normally be overlooked. If the peak value's period is very short. Let's look at a road case. Suddenly, there is a lot of traffic because of an accident, so whether that traffic is just for a few seconds, or only a few minutes, it is not going to affect any traffic, then we can tolerate this sort, but if it is for a long period of time, it would certainly affect. So at the end, always remember that effective bandwidth is the bandwidth that the network needs to allocate for the flow of traffic, the effective bandwidth is a function of three values

Traffic Profiles

A data flow should provide one of the following traffic profiles for our purposes. As you can see in the Figure 2, we have three figures: one for steady bitrate, another for volatile bitrate, and the third for bursty bitrate. I believe it is obvious from the diagram and the expression constant bitrate that it refers to a fixed rate at which time the traffic is unchanged. There has been no improvement in traffic. The traffic is moving at a really fast pace. As a result, the average data rate and peak data rate are always the same in this form of traffic.

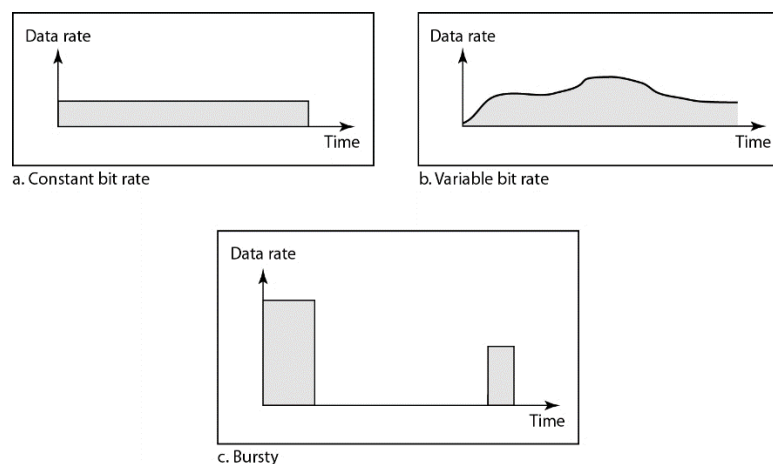


Figure 2 Three Traffic Profiles

The predictable network knows how much bandwidth to set aside for this kind of traffic ahead of time. Variable bitrate is the next choice. In this class of segment, the rate of data flow varies over time, implying that the changes are gradual rather than abrupt. There are days where there is more traffic than there is when there is less traffic. We know what we do if the timing is nine to five if we are operating with numbers, such as whether we are learning at a university or in a college. Right, but there will undoubtedly be a lot of traffic at 5 p.m. But we'll certainly have to deal with traffic, and there will be some variation: there will be a high, and then there will be a low.

As a result, rather than abrupt and sharp shifts, the data flow changes over time with gradual transitions. The average data rate and the peak data rate are both different in this form of flow. Typically, the maximal burst size is a small number. Since it does not need to be reshaped, this kind of traffic is much more difficult to manage than continuous bitrate traffic. The next one is the bursty

data rate, which gets its name from the fact that there was no traffic for a long time but then a massive influx of traffic.

As a result, the data rate varies abruptly in a brief amount of time, implying that it is jumping from zero. As an example, one Mbps can be converted to microseconds and vice versa. It's possible that it'll stay at this level for a bit. Because of the abrupt shift in traffic, average bitrate and peak data rate, or both, are different values in this situation. If you can see in the diagram, there was no data flow when there was no traffic, but suddenly there is a lot of traffic. Bursty traffic congestion in the network may occur if the demand on the network is greater than the network's capability. For instance, there is excessive traffic on the road. The road's power was smaller than the load it could handle. Congestion management refers to the mechanisms and procedures used to control congestion and maintain load below availability. Congestion is a significant problem in a packet switch network.

Network Performance

So, let's talk about network performance and how it affects it. As you can see in the diagram, there are input queues and output queues. Since routers and switches have queues buffers that carry packets before and after processing, network or inter network congestion exists.

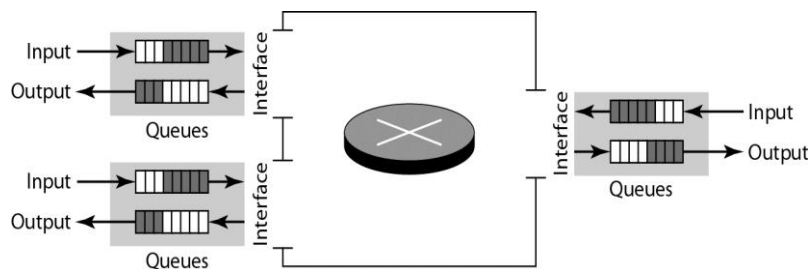


Figure 3 Queues in a Router

Each interface on a router, for example, has its own input and output queue as seen in Figure 3. When a packet is sent at the inbound terminal. There are three stages in the process. The packet is placed at the bottom of the input queue and will be reviewed later. The packet is removed from the input queue by the router's processing machine. The packet is placed in the proper output queue until it enters the front of the queue and uses its routing table and destination address to locate the source. And now it's your turn to submit. As a result, teachers, we must be mindful of two problems. If the rate of packet arrival exceeds the rate of packet processing, so indeed. The input queues are becoming increasingly long. The performance queue becomes longer and longer if the packet departure rate is lower than the packet processing rate. As a function of load throughput, packet delay and throughput indicate processor performance. So, congestion management requires two metrics that assess a network's performance: throughput and delay throughput. Throughput refers to the network's efficiency. If the power is less than the load, the throughput would inevitably be lower. When the power exceeds the load. As a result, we will almost certainly achieve high network reliability, allowing us to avoid congestion before it occurs or to alleviate congestion after it occurs as seen in Figure 4.

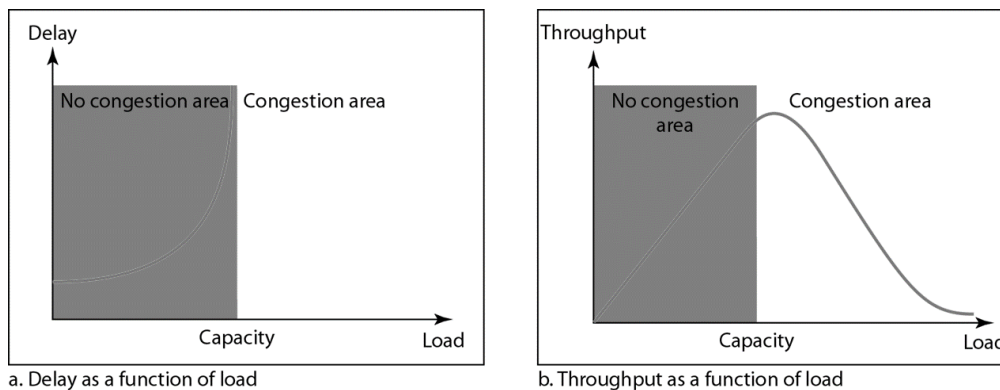


Figure 4 Packet Delay and throughput as a function of Load

11.2 Congestion Control

As a result, we can split these congestion management methods into two categories: open loop and closed loop. Open loop refers to the avoidance of congestion before it occurs. Closed loop refers to the steps that can be taken before congestion occurs.

- open-loop congestion control (prevention) and
- closed-loop congestion control (removal).

Now that we know the congestion has already happened, the next step is to figure out how to get rid of it. I understand that there is still so much traffic on the lane, but I'm not sure how to alleviate it.

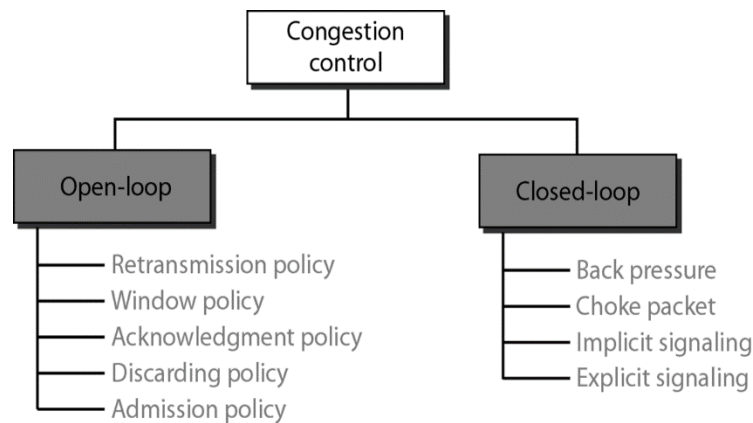


Figure 5 Categories of Congestion Control

Open Loop

To begin with, open loop, we are going to take several steps to avoid congestion and ensure that it does not exist. So, we're going to talk about open loop congestion management strategies first, and then we'll talk about other tactics. As a result, they are used to avoid congestion until it occurs. Either the source or the destination is in charge of congestion management. Retransmission policy, window policy, acknowledgment policy, discarding policy, and admission policy are some of the strategies used in open loop.

Retransmission Policy

The first is a collection of retransmission policies. Retransmission is often inevitable if the sender suspects that a packet has been destroyed or compromised, in which case the packets must be resent. If you're retransmitting the packet, this is the case. It is certain that it will increase congestion. It is likely that your packet will not meet its destination due to congestion because there is still so much traffic, and your packet will get trapped in the traffic. But, if you retransmit it, it will increase traffic, potentially increasing network congestion.

A successful retransmission strategy, on the other hand, will avoid congestion. What we should do here is plan the retransmission protocol and retransmission timers to maximise reliability while still preventing congestion. TCP's retransmission strategy, for example, is intended to avoid or reduce congestion.

Window policy

The second is the window policy. We're talking about a selective repeat window, as you can probably tell from the label. As a result, congestion can be affected in this kind of window at the sender. For congestion management, the Selective Repeat Window is preferable to the Go Back. We've also covered these two strategies in previous videos: selective repeat and go back N.

Congestion can also be affected by the type of window on the sender side. Although several packets may be successfully received at the receiver side, some packets in the Go-back-n window are resent. This duplication has the potential to worsen the network's congestion. As a result, the Selective Repeat Window can be used because it sends the same packet that was missed.

Discarding Policy

A positive discarding strategy implemented by routers is that it allows them to avoid congestion while still partly discarding compromised or less sensitive packages while maintaining message

accuracy. When transmitting audio files, routers will discard less sensitive packets to avoid congestion while maintaining the audio file's consistency.

Acknowledgment Policy

Since recognition is a component of network load, the acknowledgment policy enforced by the recipient may have an impact on congestion. Congestion caused by acknowledgment can be avoided using a variety of methods. Rather than sending acknowledgement for a single packet, the receiver should send acknowledgement for N packets. And when a packet must be sent or a timer expires should the recipient send an acknowledgment.

Admission Policy

A system should be used in admission policies to avoid congestion. Until transmitting a network flow further, switches in a flow can review the resource requirements. To avoid more congestion, the router should refuse building a virtual network link if there is a risk of congestion or if the network is still congested.

Closed Loop

In the closed loop strategy, we attempt to alleviate congestion after it has occurred. Different protocols will utilize a variety of mechanisms to accomplish their goals. So, the first is back pressure.

Back Pressure

As seen in the diagram below. Back pressure refers to a congestion control mechanism in which a congestion node blocks data from the upstream node or nodes it is connected to. Back pressure is a node to node congestion control that starts with the node and propagates in the opposite direction of a data flow to the source, back pressure is a node to node congestion control that starts with the node and propagates in the opposite direction of a data flow to the source, only a virtual circuit network in which each node knows the upstream node for whom a data flow is arriving may use the backpressure approach. Backpressure, as seen in the picture, indicates that they are moving in the other direction. Assume that node three in the Figure 6 receives more data than it can process. So, node three node three, we have congestion indicated on the diagram, which indicates it has more input data, which means capacity is smaller but demand is greater. Congestion has already happened at this time.

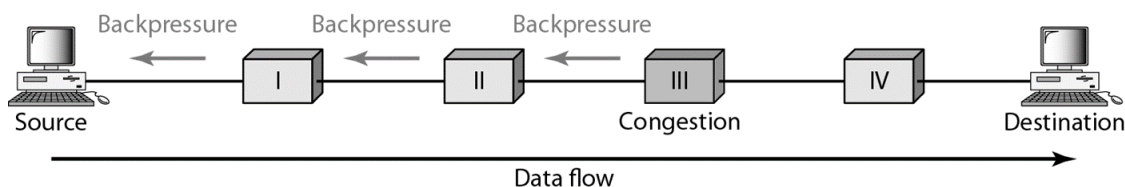


Figure 6 Backpressure

So, what it will do is drop some packets in their input buffer, and it will surely tell the node to upstream mode, which means it will notify the node instantly that it is connected to this. As a result, it will tell the node to slow down. Please don't send me any more packets, since node two will undoubtedly alert node one. If node two is congested, it notifies node one to slow down, which may cause congestion. If this is the case, node one notifies the source offer data that it should slow down. This, in turn, relieves congestion over time. It's worth noting that the pressure on node three is directed backwards to the source in order to relieve congestion; they aren't immediately informing the source. They're alerting the upstream node; node three is telling node two to shut down; node two will check to see if it's already full, and then it'll tell node one to shut down; node one will tell the source to slow down; and node one will tell the source not to transmit any more packets. Because there is already congestion on the network, they are not immediately telling the source in the event of backpressure.

Choke Packet

Choke packets are packets delivered by a node to the source warning them of congestion. They are directly telling the source of the congestion. The backpressure approach differs from the choke packet approach in that the warning is sent from one node to its upstream node. Although the warning may finally reach the source station in the choke packet, it is from the router, which has met congestion straight to the source station. The intermediary nodes, such as node second or node first, through which the packet has passed, are not being alerted here.

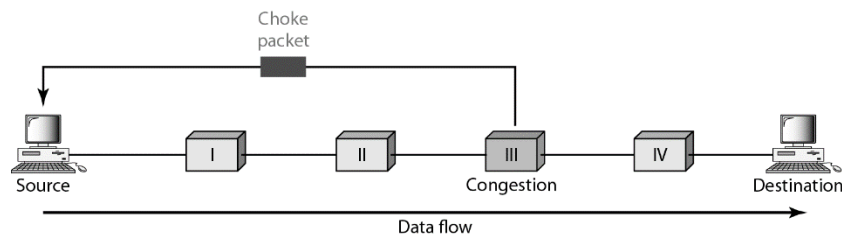


Figure 7 Choke Packet

The Figure 7 is drawn here, as you can see in the illustration. This form of control may be seen in the ICMP Internet Control Message Protocol, as you can see. When an internet router becomes overburdened with IP datagrams. It's possible that some of them will be discarded. So, it sends a source quench ICMP message (Internet Control Message Protocol) to the source code, and the warning message is sent straight to the source station, bypassing the intermediate routers. I have informed you that the ICMP packet is the lowest priority packet when I taught you about the ICMP Internet Control Message Protocol.

Implicit Signaling

If there is no communication between the crowded node and the source, who will inform about the congestion? In this instance, the source will assume that the network is crowded, especially if it has been for a long time. Because the source is not receiving any type of acknowledgement or attention, it will presume that the network is congested. When a sender sends several packets and no acknowledgement is received over an extended period of time, the source will assume that there is a congestion and that packets are not arriving.

Explicit Signaling

If a node encounters congestion, it will send a packet to the source or destination to advise them of the problem. As a result, congestion can occur as a result of the packet or the acknowledgement. As a result, it is also a responsibility to tell both the source and the destination about the congestion. As a result, explicit communication can take place in either a forward or a backward manner. When forward signalling is used, the signal is transmitted towards the direction of the congestion. The destination is alerted to the traffic. There is a source node destination node in the case of forward destination means the destination. In the case of a forward signalling receiver, use the following strategies to avoid further congestion. In the instance of backward signalling, informed sources advised that it was necessary to slow down and stop transmitting extra packets due to congestion.

11.3 Quality of Service

Quality of service (QoS) is an internetworking topic that has gotten more attention than it has been defined. Informally, we might describe quality of service as something that a flow strives towards.

Flow Characteristics

A flow has traditionally been assigned four characteristics: reliability, delay, jitter, and bandwidth as shown in Figure 8.

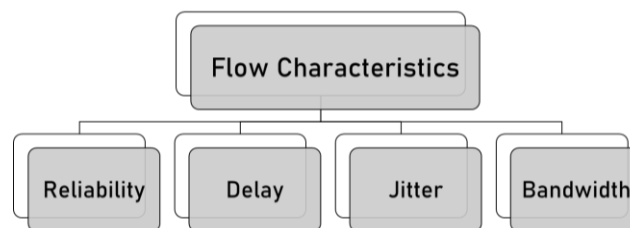


Figure 8 Flow Characteristics

Reliability

A flow requires the quality of reliability. Losing a packet or acknowledgement due to a lack of dependability necessitates retransmission. The sensitivity of application programmes to dependability, on the other hand, is not the same. Electronic mail, file transfer, and Internet connection, for example, are more crucial than phone or audio conferencing in terms of reliability.

Delay

Another flow feature is source-to-destination latency. Again, various applications may tolerate delays to varying degrees. Telephony, audio conferencing, video conferencing, and remote log-in all require minimal latency in this situation, whereas file transmission and e-mail are less critical.

Jitter

Jitter is the difference in delay between packets in the same flow. For example, if four packets leave at 0:01:02:03 and arrive at 20:21:22:23, they all have the same 20-unit delay. If the aforementioned four packets arrive at 21, 23, 21, and 28, on the other hand, they will have distinct delays: 21, 22, 19, and 24.

The first scenario is perfectly suitable for audio and video applications; the second situation is not. It makes no difference whether the packets arrive with a small or lengthy delay for these applications as long as the delay is consistent across all packets. The second example is not acceptable for this application. The variance in packet delay is referred to as jitter. High jitter denotes a considerable variance in delays; low jitter denotes a modest variance.

Bandwidth

Various applications necessitate various bandwidths. To refresh a colour screen in video conferencing, we need to transfer millions of bits every second, although the total amount of bits in an e-mail may not even approach a million.

11.4 Techniques to improve quality of service

We go through various approaches that may be utilized to improve service quality. We'll go through four typical ways briefly: scheduling, traffic shaping, admission control, and resource reservation.

Scheduling

Different flows of packets arrive at a switch or router for processing. A smart scheduling strategy balances and appropriately distributes the various flows. Several scheduling approaches have been developed to improve service quality. Three of them are discussed in this article: *FIFO queuing*, *priority queuing*, and *weighted fair queuing*.

FIFO Queuing

Packets wait in a buffer (queue) until the node (router or switch) is ready to handle them in first-in, first-out (FIFO) queuing. The queue will fill up if the average arrival rate is higher than the average processing rate, and incoming packets will be deleted. Those who have had to wait for a bus at a bus stop are familiar with a FIFO queue. Figure 9 is a conceptual representation of a FIFO queue?

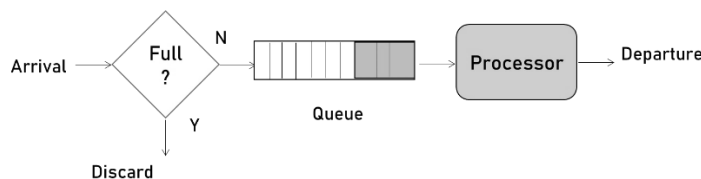


Figure 9 FIFO Queuing

Priority Queuing

Packets are initially assigned to a priority class in priority queuing. There is a separate queue for each priority class. The highest-priority queue packets are handled first. The lowest-priority queue packets are handled last. It's worth noting that the system doesn't cease servicing a queue until it's completely empty. demonstrates priority queuing with two degrees of priority. Figure 10 demonstrates priority queuing with two degrees of priority.

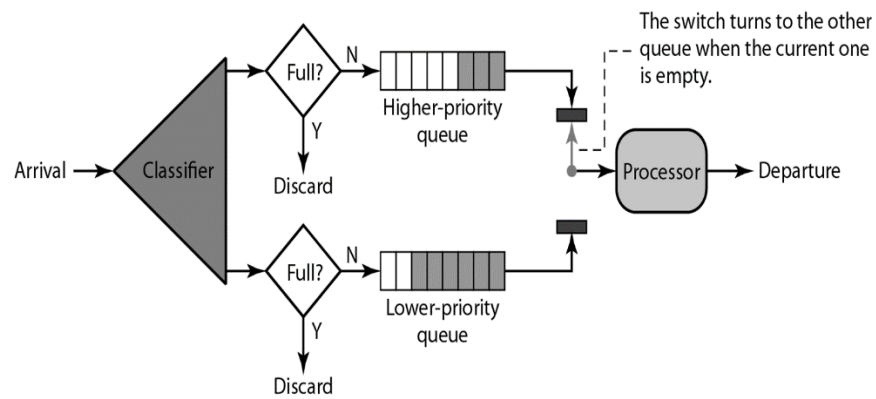


Figure 10 Priority Queuing

Because higher-priority traffic, such as multimedia, may reach its destination with less latency, a priority queue can provide better QoS than a FIFO queue. There is, however, a possible disadvantage. If a high-priority queue has a continuous flow, packets in lower-priority queues will never be processed. This is referred to as starving.

Weighted Fair Queuing

Weighted fair queuing is a superior scheduling strategy. The packets are still allocated to distinct classes and accepted to distinct queues using this method. The queues, on the other hand, are weighted according to their priority; a greater priority indicates a larger weight.

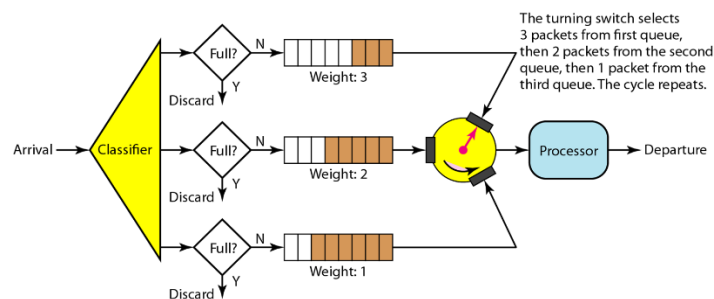


Figure 11 Weighted Fair Queuing

The system processes packets in each queue round-robin, with the number of packets chosen from each queue dependent on their weight. For example, if the weights are 3, 2, and 1, the first queue processes three packets, the second queue two, and the third queue one.

All weights can be equal if the system does not assign priority to the classes. As a result, we have fair and prioritised queuing. The methodology is shown in Figure 11 with three classes.

11.5 Traffic Shaping

Traffic shaping is a method of controlling the volume and pace of data transmitted over a network. Leaky bucket and token bucket are two strategies that may be used to shape traffic.

Leaky Bucket Algorithm

Water leaks from a bucket with a tiny hole at the bottom at a consistent pace as long as there is water in the bucket. Unless the bucket is empty, the pace at which the water leaks is unrelated to the pace at which the water is added to it. The input rate can change, but the output rate is always the same. In networking, a method known as leaky bucket helps smooth out spikes in traffic. Bursty pieces are collected in a bucket and distributed at a steady rate. Figure 12 depicts the impacts of a leaking bucket.

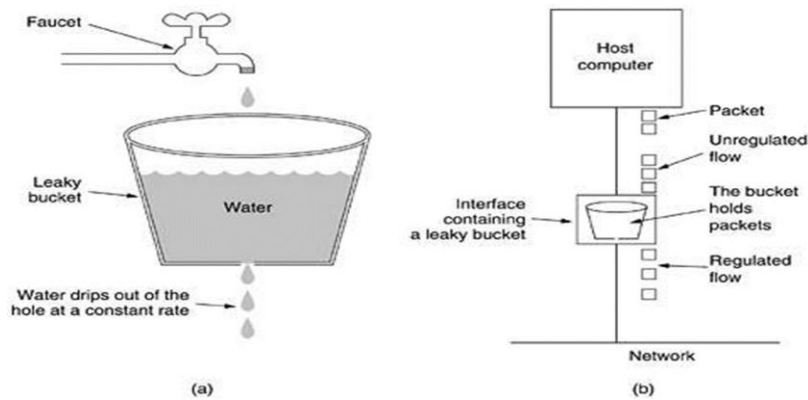


Figure 12 Leaky Bucket Algorithm

In the diagram, we suppose that the network has allocated 3 Mbps of bandwidth to a host. The leaky bucket is used to shape the input traffic so that it adheres to the commitment. Figure 12 shows a total of 24 Mbits of data, the host delivers a burst of data at a rate of 12 Mbps for 2 seconds. After 5 seconds of silence, the host delivers data at a rate of 2 Mbps for 3 seconds, totaling 6 Mbits of data. The host has transmitted 30 Mbits of data in 10s in total. The leaky bucket smooths traffic by transmitting data at a 3 Mbps rate over the same 10 seconds. The first burst may have harmed the network if it hadn't been for the leaky bucket, since it would have consumed more bandwidth than was allocated to this server. We can also understand how the leaking bucket may help to avoid congestion. Consider the motorway at rush hour as an example (bursty traffic). Congestion on our highways may be eliminated if commuters simply stagger their working hours.

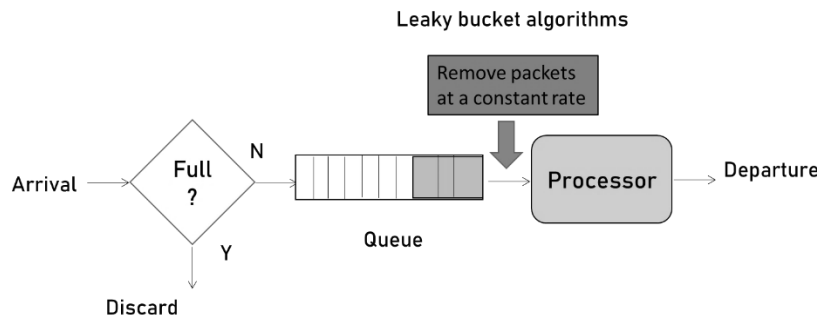


Figure 13 Leaky Bucket Implementation

Figure 13 depicts a simple leaky bucket implementation. The packets are stored in a FIFO queue. If the traffic is made up entirely of fixed-size packets. At each tick of the clock, the operation eliminates a set number of packets from the queue. If the traffic is made up of packets of varying lengths, the fixed output rate must be determined by the amount of bytes or bits.

Token Bucket

Having a leaking bucket is really restricting. An idle host is not credited. For example, if a host does not send for an extended period of time, its bucket will become empty. The leaky bucket now only supports an average rate if the host has bursty data. The duration of the host's inactivity is not taken into account. The token bucket method, on the other hand, allows idle hosts to save credit for the future in the form of tokens. The system transfers n tokens to the bucket for each tick of the clock. For each cell (or byte) of data supplied, the system eliminates one token. The bucket accumulates 10,000 tokens if n is 100 and the host is idle for 100 ticks. The host can now eat all of these tokens in a single tick with 10,000 cells, or 1000 ticks with 10 cells each. In other words, as long as the bucket isn't empty, the host can deliver bursty data. Figure 14 demonstrates the concept.

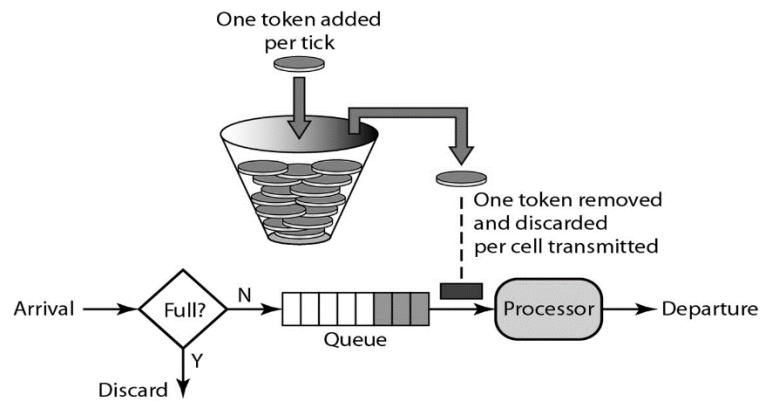


Figure 14 Token Bucket Algorithm

A counter may simply be used to implement the token bucket. The token's value is set to zero. The counter is increased by one each time a token is inserted. The counter is decremented by one each time a unit of data is transmitted. The host is unable to transfer data when the counter is 0.

Combining token bucket and leaky bucket

Both strategies can be used together to credit an idle host while also regulating traffic. After the token bucket, the leaky bucket is used; the rate of the leaky bucket must be greater than the rate of tokens dropped in the bucket.

Resource Reservation

A data flow necessitates the use of resources such as a buffer, bandwidth, and CPU time, among others. When these resources are booked ahead of time, the quality of service is increased. In this part, we'll look at one QoS model called Integrated Services, which relies significantly on resource reservation to improve service quality.

Admission Control

Admission control is the process by which a router or switch accepts or rejects a flow based on established parameters known as flow requirements. Before accepting a flow for processing, a router examines the flow specifications to verify if the router's capacity (bandwidth, buffer size, CPU speed, and so on) and past commitments to other flows are sufficient to handle the incoming flow.

Summary

- When there are too many packets in one region of the network, the subnet's performance suffers. As a result, a network's communication channel is considered crowded if packets transiting the path incur delays that are much greater than the way's propagation delay. When packets never reach their destination, the delay approaches infinite, this is referred to as extremely congested.
- The terms "congestion control" and "flow control" are not interchangeable. Flow control is concerned with point-to-point traffic between a particular source host and a specific destination host, whereas congestion is a global phenomena affecting all hosts, all routers, the store-and-forward processing within the routers, and so on.
- The computer network, which is also a system, is separated into two categories according to control theory. There are two types of solutions: open loop and closed loop.
- The traffic management capability helps you to make the most of your network's resources while also ensuring that resources that haven't been expressly assigned are used efficiently. The majority of traffic management will be determined by transmit priority and bandwidth availability. Delay-sensitive traffic is given a higher transmit priority in the transmit priority.

- The leaky bucket technique is used in network traffic shaping and rate limiting applications. The technique allows you to manage the rate at which data is injected into a network, allowing you to handle data rate burstiness.

Keywords

Congestion: When packets crossing a network's communication channel incur delays that are much greater than the path's propagation delay, the channel is said to be congested.

IP Address: An Internet Protocol address (IP address) is a numerical label assigned to each device (e.g., computer, printer) participating in a computer network that uses the Internet Protocol for communication.

IP Protocol: The Internet Protocol Suite's datagrams (packets) are relayed across an internetwork using the Internet Protocol Suite's Internet Protocol (IP).

Traffic Shaping: Attempts to normalise the average data transfer rate.

Review Questions

1. Explain the general principles of congestion.
2. What do you understand by QoS? Describe the basic QoS structure.
3. Discuss the following two algorithms:
 - a. Leaky Bucket
 - b. Token Bucket
4. What are two types of congestion control? Where is congestion control implemented in each case?
5. Explain all traffic shaping techniques.
6. Write down techniques to improve quality of service.
7. Difference between token bucket and leaky bucket algorithm.

Self-Assessment

1. The technique in which a congested node stops receiving data from the immediate upstream node or nodes is called as
 - a. Explicit signalling
 - b. Back pressure
 - c. Implicit signalling
 - d. Redundant signals
2. A leaky bucket algorithm shapes bursty traffic into fixed-rate traffic by averaging the
 - a. Data rate
 - b. Average rate
 - c. Traffic rate
 - d. Traffic shaping
3. Two classes of services have been defined for
 - a. Integrated services
 - b. Quality data services
 - c. Technical services
 - d. Protocol services
4. In open-loop congestion control, policies are applied to
 - a. Prevent congestion
 - b. Discard congestion
 - c. Maximize congestion

- d. Eliminate congestion
- 5. A mechanism to control the amount and the rate of the traffic sent to the network is called
 - a. Traffic congestion
 - b. Traffic flow
 - c. Traffic control
 - d. Traffic shaping
- 6. Scheduling is done by
 - a. Weighted fair queuing
 - b. FIFO
 - c. Random
 - d. LIFO
- 7. Which of the following is a congested control algorithm.
 - a. The leaky bucket
 - b. Token bucket
 - c. Resource reservation
 - d. All of above
- 8. In Congestion Control, the packet is put at the end of the input queue while waiting to be
 - a. checked
 - b. entered
 - c. read
 - d. interpret
- 9. Integrated Services is based on flow based Quality OF Service model designed for
 - a. CPU
 - b. Data Node
 - c. IP
 - d. Traffic Shaping
- 10. The token bucket can easily be implemented with a counter, initialized by
 - a. 0
 - b. 1
 - c. -1
 - d. -2
- 11. In Congestion, to define the maximum data rate of the traffic we use
 - a. Average Data Packet
 - b. Peak Data Rate
 - c. Packet Data Rate
 - d. Average Data Rate
- 12. In Congestion, the maximum burst size normally refers to the maximum length of time the traffic is generated at the
 - a. Average Rate
 - b. Packet Rate
 - c. Protocol Rate
 - d. Peak Rate

Self-Assessment Answer

1.	b	2.	a	3.	a	4.	a
5.	d	6.	d	7.	d	8.	a
9.	c	10.	a	11.	b	12.	d

Further Readings

Achyut S Godbole and Atul Kahate, *Web Technologies*, Tata McGraw Hill.

Andrew S. Tanenbaum, *Computer Networks*, Prentice Hall.

Behrouz A. Forouzan, Sophia Chung Fegan, *Data Communications and Networking*, McGraw-Hill Companies.

Douglas Comer, *Computer Networks and Internets with Internet Applications*, 4th Edition, Prentice Hall.

Ferguson P., Huston G., John Wiley & Sons, Inc., 1998. *Quality of Service: Delivering QoS on the Internet and in Corporate Networks*.

J. D. Spragins, *Telecommunications Protocols and Design*, Addison Wesley.

McDysan, David E. and Darren L. Spohn, *ATM Theory and Applications*, McGraw-Hill Osborne Media, 1998.

Nassar, Daniel J., *Ethernet and Token Ring Optimization*, iUniverse.com, 2000.

Spurgeon, Charles E. *Ethernet, The Definitive Guide*. O'Reilly & Associates, 2000.

William A Shay, *Understanding Communication and Networks*, 3rd Edition, Thomson Press.